



PORTFOLIO

PRODUCT DESIGN ENGINEERING

Presented by :
Chethan Nittur Kanakamurthy

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SUMMARY

I am a Mechanical Design Engineer focused on building strong fundamentals in product design, manufacturing, and system-level thinking. My work reflects a hands-on approach to understanding how real products are designed, built, and scaled under practical constraints.

My long-term goal is to grow into a leadership role, with the ambition of becoming a CEO, by learning how different functions engineering, manufacturing, supply chain, and operations work together to create reliable products. This portfolio represents my learning in progress and my commitment to mastering engineering as the foundation for that journey.



[Chethan N K](#)

PROJECTS

- Motion Multiplication Mechanism for Top Latch in an Exit Device
- Low Profile Deadbolt – Doublr Crank Mechanism
- Electronic Lock Drive Train. (Upcoming)

This portfolio is maintained as a live engineering record. Projects documented here represent ongoing work, and each section will be expanded with deeper analysis, testing, and iteration as designs mature. This format reflects my belief that strong engineering is an iterative process, refined over time through validation and learning.

PROJECT – 1

MOTION MULTIPLICATION MECHANISM FOR TOP LATCH IN AN EXIT DEVICE

DOMAIN

LIFE-SAFETY HARDWARE | ELECTROMECHANICAL / MECHANICAL SYSTEMS

CONTEXT

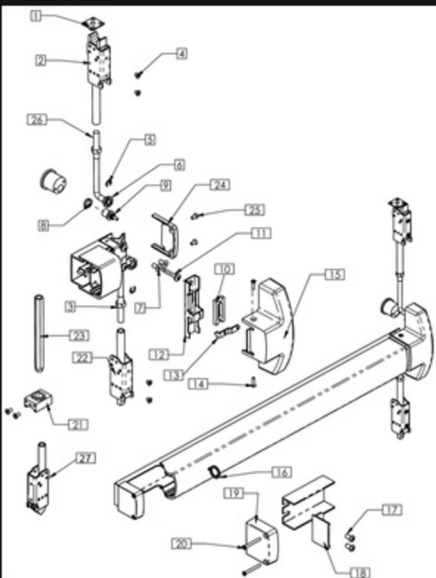
THE EXISTING EXIT DEVICE DESIGN PROVIDED APPROXIMATELY 1 INCH OF TOP LATCH ENGAGEMENT WITH THE DOOR FRAME. UNDER WINDSTORM AND TORNADO CONDITIONS, THIS LIMITED ENGAGEMENT RESULTED IN INSUFFICIENT LATCH SURFACE CONTACT, LEADING TO INSTANCES OF PARTIAL DISENGAGEMENT AND UNLATCHING DUE TO FLUCTUATING LOADS, DOOR DEFLECTION, AND FRAME MOVEMENT.

THE ISSUE WAS NOT USER ACTUATION, BUT INSUFFICIENT MECHANICAL ENGAGEMENT LENGTH TO RELIABLY RESIST EXTERNAL FORCES.

MY ROLE

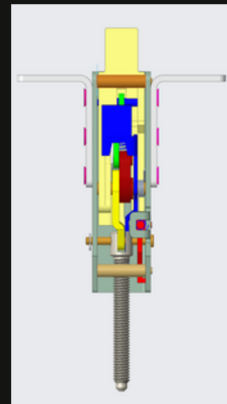
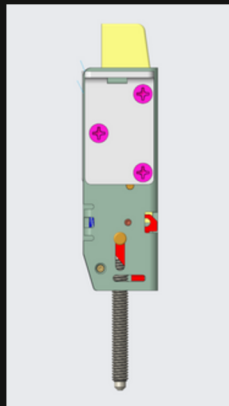
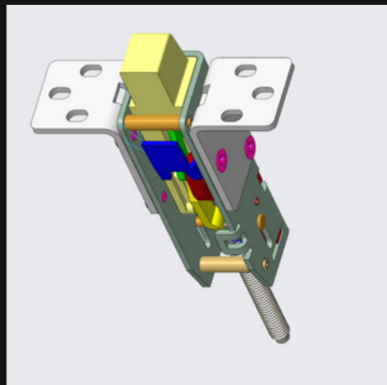
- CONCEPT GENERATION & MECHANISM DESIGN
- MOTION MULTIPLICATION STRATEGY
- KINEMATIC REASONING & PACKAGING OPTIMIZATION
- DFM-AWARE DESIGN THINKING

DUE TO CONFIDENTIALITY, PROPRIETARY DIMENSIONS AND DETAILED GEOMETRIES ARE OMITTED. ILLUSTRATIONS ARE REPRESENTATIVE.



DESIGN OBJECTIVE

- INCREASE TOP LATCH ENGAGEMENT FROM 1" TO 2"
- PROVIDE ADDITIONAL LATCH SURFACE AREA FOR LOAD RESISTANCE
- MAINTAIN EXISTING PUSH-BAR ERGONOMICS
- PRESERVE PACKAGING, RELIABILITY, AND MANUFACTURABILITY



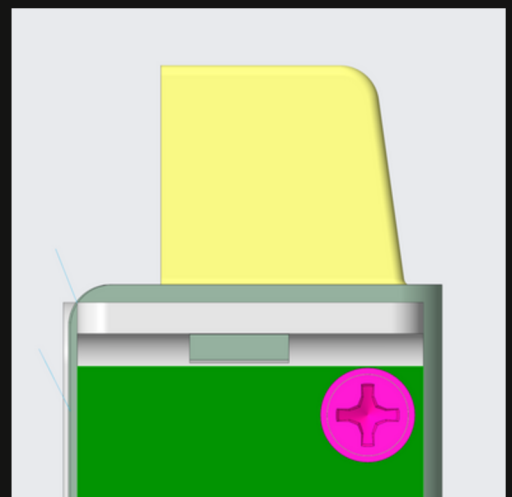
NOTE: CAD IMAGES SHOWN ARE REPRESENTATIVE AND FOR REFERENCE ONLY, DIMENSIONS ARE CONFIDENTIAL

SELECTED MECHANISM

A CAM-DRIVEN MOTION MULTIPLICATION MECHANISM WAS DESIGNED TO TRANSFORM LIMITED INPUT MOTION INTO INCREASED LINEAR TRAVEL OF THE TOP LATCH. THE CAM PROFILE WAS SHAPED TO MAXIMIZE LATCH DISPLACEMENT DURING THE ENGAGEMENT PHASE, WHILE MAINTAINING SMOOTH ACTUATION AND RETURN.

KEY DESIGN FOCUS AREAS:

- CAM PROFILE GEOMETRY
- FOLLOWER PATH CONTROL
- LOAD TRANSFER AND CONTACT STRESSES
- RETURN SPRING COMPATIBILITY



KEY RESULT

- TOP LATCH ENGAGEMENT INCREASED FROM 1" TO 2"
- 100% INCREASE IN ENGAGED LATCH SURFACE
- NO CHANGE TO EXTERNAL FORM FACTOR
- NO INCREASE IN PUSH-BAR STROKE OR PERCEIVED USER EFFORT
- IMPROVED RESISTANCE TO WIND-INDUCED UNLATCHING

IMPACT & VALUE

BY ACHIEVING ADDITIONAL LATCH ENGAGEMENT WITHOUT CHANGING THE PRODUCT FORM FACTOR, THIS DESIGN ADDRESSED THE ROOT CAUSE OF WIND-RELATED UNLATCHING WHILE REMAINING COMPATIBLE WITH EXISTING INSTALLATION, TOOLING, AND USER INTERACTION. THE PROJECT DEMONSTRATED HOW KINEMATIC OPTIMIZATION WITHIN FIXED CONSTRAINTS CAN DELIVER SUBSTANTIAL SAFETY IMPROVEMENTS.



CAM

PROJECT – 2

LOW PROFILE DEADBOLT – USING DOUBLE CRANK MECHANISM

DOMAIN

RESIDENTIAL HARDWARE | COMPETITIVE PRODUCT DESIGN | SCHLAGE

CONTEXT

THE Kwikset DOWNTOWN SERIES HAS GAINED SIGNIFICANT MARKET SHARE BY OFFERING A MODERN, LOW-PROFILE AESTHETIC THAT APPEALS TO URBAN HOMEOWNERS. HOWEVER, ACHIEVING THIS SLIM PROFILE OFTEN COMES WITH TRADE-OFFS IN INTERNAL MECHANISM SPACE. THE CHALLENGE WAS TO DEVELOP A SCHLAGE COMPETITOR PRODUCT THAT MATCHES THE 0.5-INCH PROTRUSION OF THE COMPETITION BUT EXCEEDS THEIR PERFORMANCE BY MAINTAINING BHMA GRADE 1 SECURITY STANDARDS. THE GOAL WAS TO MARRY KWIKSET'S MINIMALIST APPEAL WITH SCHLAGE'S REPUTATION FOR ROBUST, COMMERCIAL-GRADE DURABILITY.

MY ROLE

- COMPETITIVE BENCHMARKING (SCHLAGE VS. KWIKSET DOWNTOWN)
- MECHANISM SYNTHESIS & KINEMATIC ANALYSIS
- SPACE CLAIM OPTIMIZATION & 3D MODELING
- STRUCTURAL SIMULATION (FEA) FOR BHMA GRADE 1 LOADS
- PROTOTYPING & TOLERANCE ANALYSIS

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KWIKSET DOWNTOWN SERIES



DESIGN OBJECTIVE

- MATCH THE 0.5-INCH LOW-PROFILE AESTHETIC OF THE KWIKSET DOWNTOWN SERIES.
- SURPASS COMPETITOR DURABILITY BY ACHIEVING BHMA GRADE 1 CERTIFICATION.
- MAINTAIN STANDARD 1-INCH DEADBOLT THROW WITHIN THE REDUCED HOUSING.
- ENSURE ZERO-COMPROMISE ON SCHLAGE'S ROBUST METAL CONSTRUCTION.

SELECTED MECHANISM

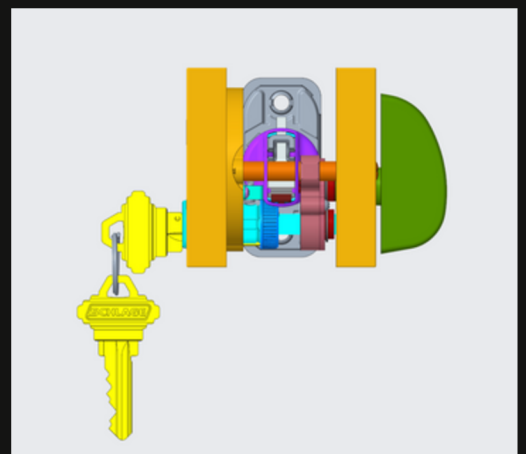
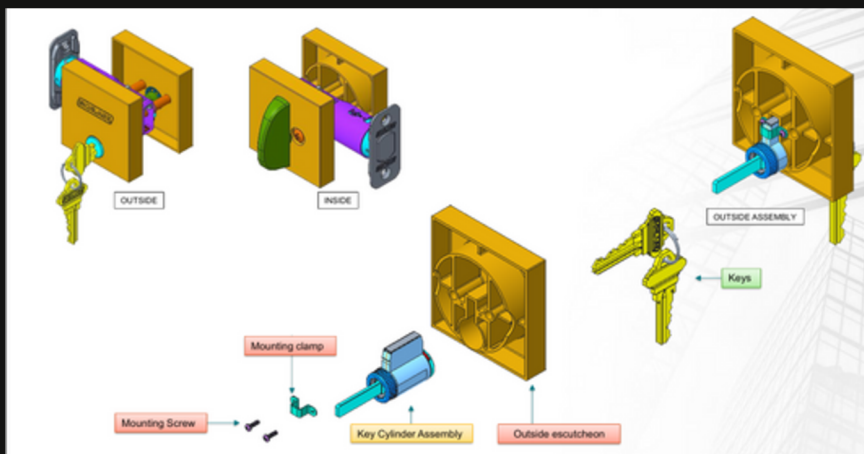
DOUBLE CRANK MECHANISM WAS SELECTED TO TRANSLATE THE ROTATIONAL INPUT FROM THE KEY/THUMB-TURN INTO THE LINEAR MOTION REQUIRED TO DRIVE THE BOLT. UNLIKE TRADITIONAL RACK-AND-PINION SYSTEMS USED IN BULKIER LOCKS, THE DOUBLE CRANK LINKAGE OPERATES ON A FLAT PLANE. THIS ALLOWED THE MECHANISM TO "FOLD" INTO THE THIN 0.5-INCH HOUSING, ENABLING SCHLAGE TO MATCH THE COMPETITOR'S FORM FACTOR WITHOUT SACRIFICING THE MECHANICAL ADVANTAGE NEEDED FOR A HEAVY-DUTY GRADE 1 LATCH.

TO REALIZE THIS COMPACT DESIGN WITHOUT SACRIFICING STRENGTH:

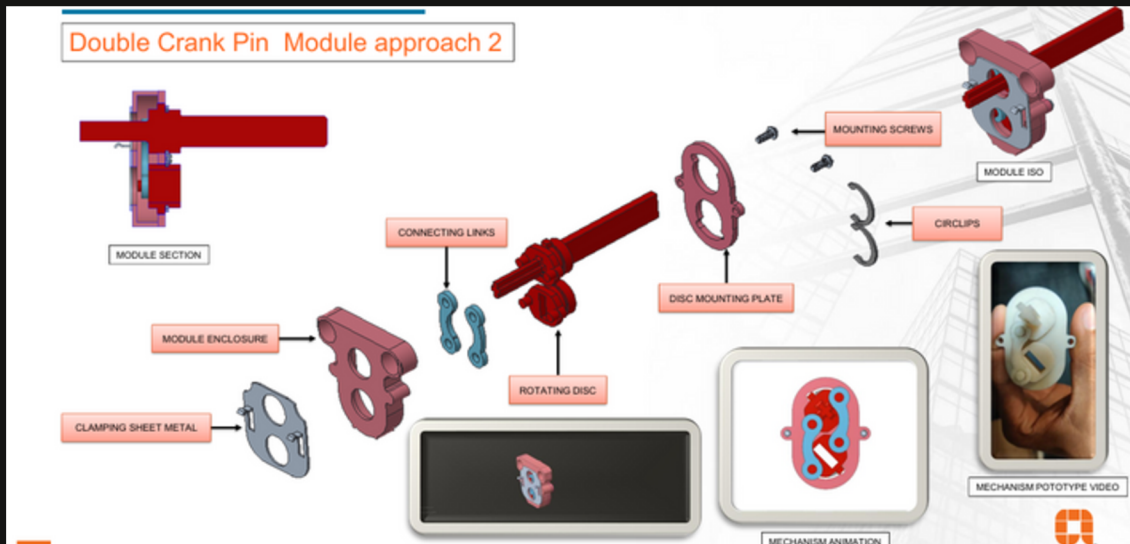
- PRESSURE DIE CASTING (PDC): USED FOR THE ESCUTCHEON (HOUSING) TO ACHIEVE COMPLEX INTERNAL GEOMETRY AND INTEGRAL MOUNTING BOSSES WITHIN A THIN PROFILE, PROVIDING RIGID SUPPORT FOR THE MECHANISM.
- POWDER METALLURGY (PM): EMPLOYED FOR THE CRANK ARMS TO DELIVER HIGH-STRENGTH, NET-SHAPE COMPONENTS CAPABLE OF HANDLING HIGH TORQUE WITHOUT MACHINING.
- SHEET METAL: UTILIZED FOR INTERNAL RETAINING PLATES TO MINIMIZE STACK-UP HEIGHT.

KEY DESIGN FOCUS AREAS:

- LINKAGE GEOMETRY: OPTIMIZING LINK LENGTHS TO PREVENT KINEMATIC SINGULARITIES (LOCKING UP) WHILE MAXIMIZING THROW.
- SHEAR STRENGTH: DESIGNING PIVOT POINTS TO WITHSTAND FORCED ENTRY LOADS (KICK-IN RESISTANCE) SUPERIOR TO STANDARD RESIDENTIAL LOCKS.
- COMPONENT NESTING: STACKING COMPONENTS EFFICIENTLY TO ELIMINATE WASTED INTERNAL AIR SPACE.



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KEY RESULT

- **SUCCESSFULLY ACHIEVED 0.5-INCH PROFILE TO COMPETE DIRECTLY WITH KWIKSET.**
- **VALIDATED MECHANISM STRENGTH AGAINST GRADE 1 IMPACT STANDARDS (HIGHER THAN STANDARD RESIDENTIAL).**
- **SUCCESSFUL INTEGRATION OF FULL 1-INCH THROW WITHIN COMPACT FOOTPRINT.**
- **PRESERVED SCHLAGE'S PREMIUM TACTILE FEEL AND KEY TORQUE REQUIREMENTS.**

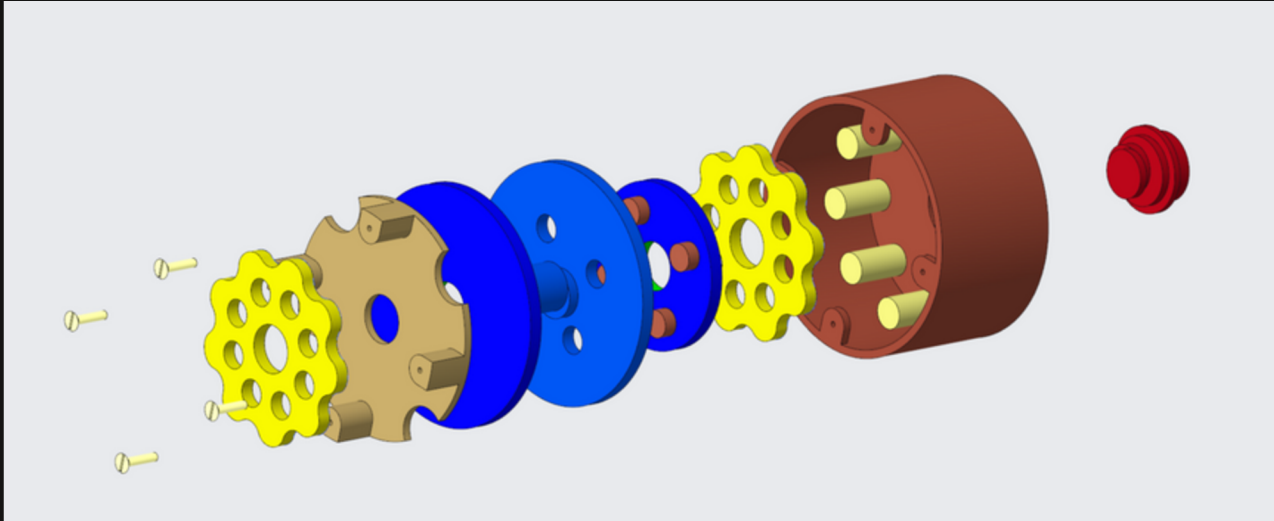
IMPACT & VALUE

THIS DESIGN CLOSSES A CRITICAL GAP IN SCHLAGE'S PORTFOLIO, ALLOWING THE BRAND TO COMPETE IN THE MODERN/URBAN SEGMENT PREVIOUSLY DOMINATED BY THE KWIKSET DOWNTOWN SERIES. BY DELIVERING A "NO COMPROMISE" SOLUTION—SLEEK LOOKS WITH GRADE 1 SECURITY—THE PRODUCT PREVENTS CUSTOMER MIGRATION TO COMPETITORS SOLELY BASED ON AESTHETICS, REINFORCING SCHLAGE AS THE LEADER IN BOTH SAFETY AND DESIGN.

High Level Requirements	Criteria Characteristics (Should Have / Nice to Have)	KWIKSET Baseline	Concept 4 – MODULAR
1 Meet Res Grade 1 / AAA in-line with ANSI/BHMA A156.36 / A156.40	Should Have	Yes	Yes – Latch modification to be verified
2 Accommodates Standard 2-1/8" without re-prepping	Should Have	Yes	Yes
3 Accommodates standard 1-3/8" to 1-3/4" Door thickness without any modifications	Should Have	1 3/4" is Standard – Have kit for 1 3/8"	
4 Accommodates Thick Door up to 2-1/4" with additional thick door kit	Should Have	Yes	Yes
5 Angle of retraction of Key and Thumb Turn to meet current Standard Deadbolt Specifications	Should Have	No ≈ 120 Deg AOR	Yes – Meets Current 90 Deg AOR
6 Force Applied on Key and Turn to meet current Standard Deadbolt Specifications	Should Have	Yes	Yes
7 ADA compliant thumb turn design	Should Have	Yes	Yes
8 Differentiating features with respect to Kwikset IP Aspects	Nice to Have	No	Yes – Advantage of In-Line approach
9 Ease of Assembly for the Installer (3 Piece)	Should Have	Yes	Yes-If the module is in the inside assembly
10 Deadbolt and Cylinder: Fully compatible with Schlage 5 pin offering	Should Have	Yes - for 5 Pin	Yes
11 Deadbolt and Cylinder: Compatible with Schlage 6 Pin offering	Nice to Have	No 6 Pin Offering	No 6 Pin offering
12 UL 10C 20 minutes for wood door	Should Have	No – Optional UL 20 Mins Rating	To be Reviewed during detail design
13 Lower BOM Cost	Nice to Have	-	Yes
14 Rose Profile - 0.5" or lesser	Should Have	Yes	Yes
15 Lesser Modification to Latch Components	Should Have	-	Yes - Secondary Operations on Latch housing
16 Screws: Retain Schlage-standard 1/4" mounting thru-bolts	Nice to Have	-	Yes – Screw length changed
17 Lesser Part Count	Nice to Have	-	No-because of mechanism
18 UL 10C Rating - 90 minutes – UL Extended Rating - 3 hours with fire cup	Nice to Have	No	No – To be Reviewed during detail design
19 Accommodates 1-1/2" Door bore without re-prepping	Nice to Have	No	Yes-Because of module
20 Chassis Robustness: Perception of quality/security	Nice to Have	Yes	Yes

PROJECT - 1

ELECTRONIC LOCK DRIVE TRAIN. (UPCOMING)



UPCOMING.....